

Suman Behera

Data Scientist | Machine Learning Engineer

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SUMMARY

Data Scientist and Machine Learning Engineer who builds and deploys ML systems using Scikit-learn, PyTorch, FastAPI, Docker, MLflow, and DVC. Delivered recommendation, NLP, and MLOps projects handling datasets with millions of records, with cloud inference services deployed on Render and Hugging Face Spaces.

SKILLS

Languages:	Python, SQL
Data Science & ML:	Machine Learning, Data Analysis, Data Cleaning, EDA, Feature Engineering, Predictive Modeling, Classification, Regression, Clustering, Supervised & Unsupervised Learning, Cross-validation, Hyperparameter Tuning, A/B Testing, Hypothesis Testing, Statistical Analysis
Libraries & Frameworks:	Scikit-learn, Pandas, NumPy, SciPy, PyTorch, TensorFlow, Keras, Matplotlib, Seaborn
ML Algorithms:	Decision Tree, Random Forest, Gradient Boosting, XGBoost, Neural Networks, GRU, Transformer Fine-tuning, Collaborative Filtering, NCF, GRU4Rec
NLP & Transformers:	HuggingFace, DistilBERT, RoBERTa, Tokenization, Text Classification
MLOps & Cloud:	MLflow, DVC, Docker, FastAPI, Streamlit, GitHub Actions, CI/CD, Render, Hugging Face Spaces
Visualization & BI:	Power BI, Matplotlib, Seaborn, Excel
Databases & Tools:	MySQL, MongoDB, Git, VS Code, Jupyter Notebook, Google Colab, Linux

EXPERIENCE

Machine Learning Portfolio Projects | Independent Projects

Jan 2025 – Present | Remote

- Designed and delivered three containerized ML systems — an MLOps pipeline, an NLP classifier, and a hybrid recommender — each with deployed APIs on Render and Hugging Face Spaces, handling datasets of 2.75M+ records.
- Used DVC for data versioning, MLflow for experiment tracking, Docker for containerization, and GitHub Actions for CI/CD; resolved deployment bottlenecks including Docker image optimization (11.5 GB → 434 MB) and Render memory limitations.

PROJECTS

Customer Churn Prediction — MLOps Pipeline | [GitHub](#)

Tech: Python, Scikit-learn, FastAPI, Docker, MLflow, DVC, GitHub Actions, Render

- Built a 7-stage MLOps pipeline covering ingestion, preprocessing, training, tracking, serving, containerization, and deployment; used DVC for reproducible pipeline orchestration.
- Trained a SMOTEENN + GradientBoosting model on imbalanced telecom churn data; applied resampling after the train-test split to prevent data leakage, achieving Recall: 0.813, F1: 0.607, ROC-AUC: 0.837.
- Deployed a FastAPI prediction API on Render via Docker; tracked experiments with MLflow and validated the pipeline using GitHub Actions CI and integration tests.

Hybrid E-Commerce Recommendation System | [GitHub](#)

Tech: Python, PyTorch, FastAPI, Streamlit, MLflow, Docker, Render

- Built a hybrid recommender on 2.75M+ user interactions using Item-CF, NCF, and GRU4Rec; engineered event-weighted implicit feedback (view=1, add-to-cart=3, transaction=5) and benchmarked models using HR@10 and MSE with MLflow tracking.
- Replaced a 774 MB item-similarity matrix with GRU4Rec + popularity fallback, reducing Docker image from 11.5 GB to 434 MB and enabling Render deployment.

Fake News Detection using Transformers | [GitHub](#)

Tech: Python, PyTorch, HuggingFace Transformers, RoBERTa, MLflow, FastAPI, Docker, Hugging Face Spaces

- Fine-tuned Weighted RoBERTa for fake news detection on the LIAR dataset, improving F1 by 65.5% (0.3423 → 0.5664) over a TF-IDF Logistic Regression baseline.
- Benchmarked five models (Naive Bayes, DistilBERT, RoBERTa, Weighted RoBERTa) and tracked accuracy, precision, recall, F1, and ROC-AUC using MLflow.
- Handled a Render memory limitation loading RoBERTa; migrated to Hugging Face Spaces with Docker and deployed the model via Hugging Face Hub as a FastAPI prediction API.

EDUCATION

Master of Computer Applications (MCA)

Gandhi Institute of Engineering and Technology (GIET), Gangapada, Bhubaneswar | **CGPA: 7.61 / 10**

2022 – 2024

Bachelor of Science

Divine Degree College, Nayagarh, Odisha | **CGPA: 8.36 / 10**

2019 – 2022

CERTIFICATIONS

- HackerRank SQL — [Basic](#), [Intermediate](#), [Advanced](#)
- [freeCodeCamp Machine Learning with Python](#)
- [freeCodeCamp Data Analysis with Python](#)
- [Kaggle Intro to Machine Learning](#)